

CYBER SHIELD

An Interactive Cybersecurity Awareness & Phishing-Simulation Portal

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Awareness material is passive — attacks aren't

- Phishing, SMS/UPI fraud, fake job offers, and AI-voice/deepfake scams drive a large share of individual financial fraud
- Most awareness content is read once (a poster, a PDF, a slide deck) and forgotten within days
- Generic Western/enterprise material rarely reflects everyday UPI, SMS, and social-media scams common locally
- There is no easy way to test whether the advice was actually absorbed — no feedback loop for the learner

Two options today — neither fits an individual learner

Static awareness content

- Posters, PDFs, slide decks distributed once
- No interactivity, no way to verify understanding
- Goes stale as scam tactics evolve

Enterprise phishing-simulation tools

- Effective, but built for organizations
- Needs licensing, email infrastructure, admin setup
- Inaccessible to individual learners, students, small teams

A single-file, self-serve awareness portal

- No install, no backend, no signup beyond a lightweight email session
- Scam recognition taught as an interactive “find the clue” exercise, not a static read
- Covers Email, SMS/WhatsApp, Website/URL, Social Media, AI/Deepfake, and Public Wi-Fi scams
- Practical tools (URL / password / email checkers) usable independently of the lessons
- Free-text practice arena closes the gap between recognizing and articulating the right response

What the system sets out to do

Interactive learning

Case-based modules covering the most common scam categories

Practical tools

Always-available, offline URL / password / email analyzers

Reinforcement

Checklists, short-form content, and scored free-text practice

Progress tracking

Per-learner progress and achievements to sustain engagement

Zero infrastructure

Deployable as a single static file at no hosting cost

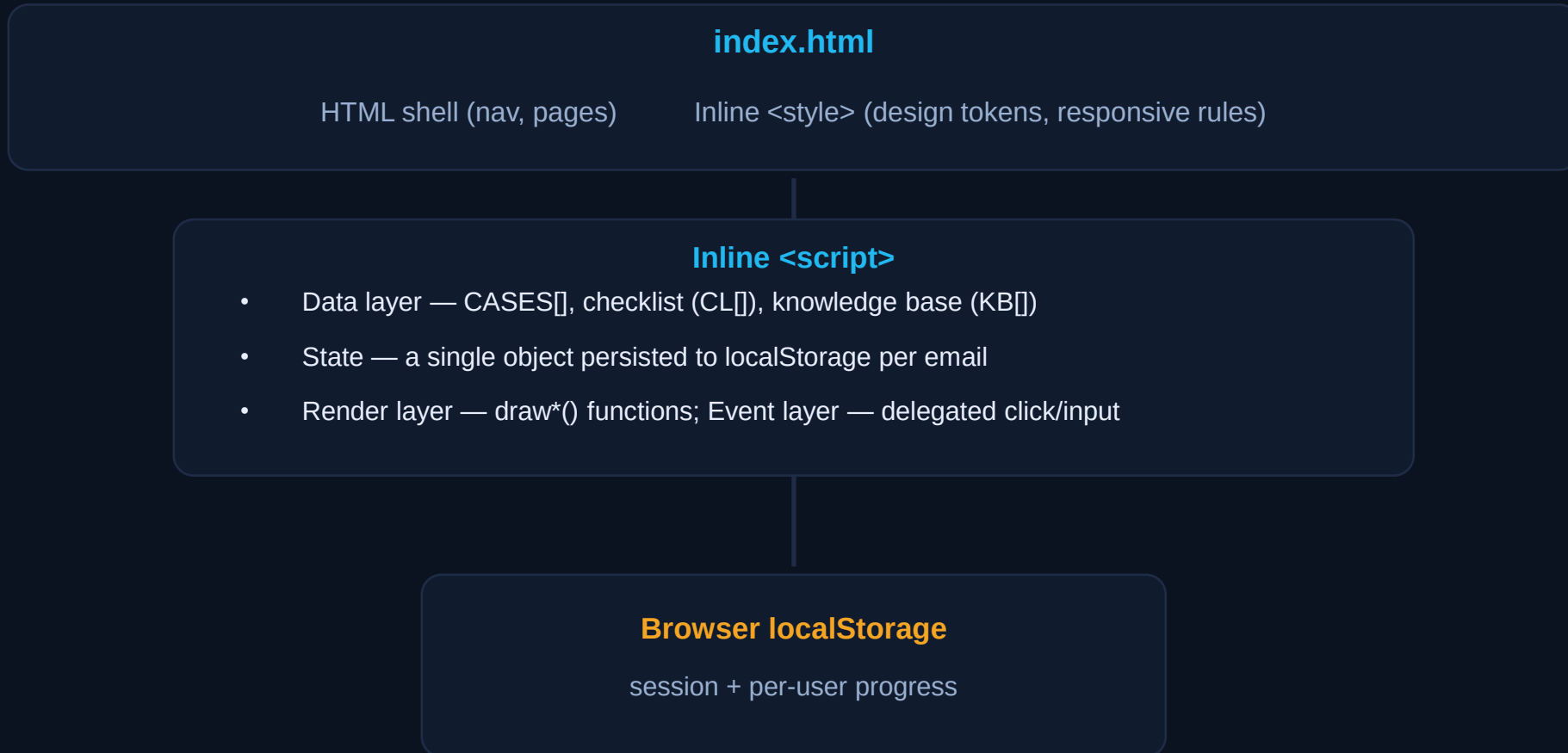
Basic security hygiene

Input validation and safe DOM insertion throughout

A deliberately zero-dependency stack

Structure	HTML5
Styling	CSS3 — custom properties, flexbox/grid, media queries, prefers-color-scheme
Behavior	Vanilla JavaScript (ES6+) — no framework, no build step
Graphics	Inline SVG — icons, bar/donut charts, procedural background mesh
Persistence	Browser localStorage, namespaced per logged-in email
Fonts	Google Fonts (Inter) — the only external network dependency

State → render, all in one page



Seven modules, one continuous experience



Overview



Investigation Lab



Safety Center



Safety Tools



Cyber Shorts



AI Practice Arena

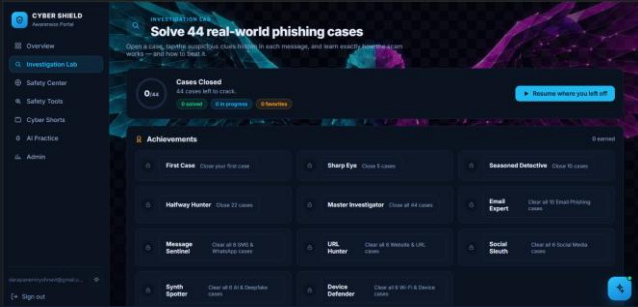


Admin Dashboard

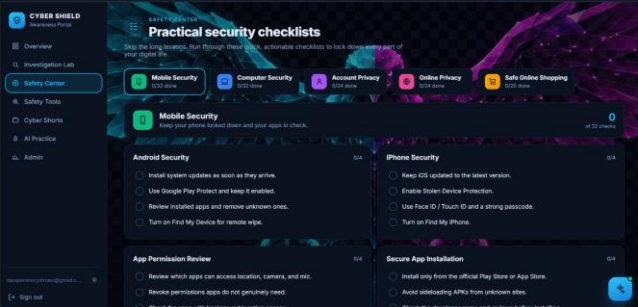
Application in action



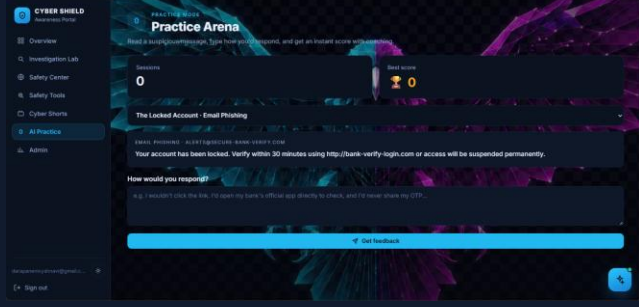
Overview — Cyber Score & quick actions



Investigation Lab — tap-the-clue cases



Safety Center — 136-item checklists



AI Practice Arena — scored free-text feedback

Replace each box with a real screenshot exported from the running app.

What we'll walk through

- Sign in and land on the Overview — Cyber Score, level, quick actions
- Open an Investigation Lab case, tap clues, unlock the safest move
- Run the URL and email safety tools on a live example
- Complete a checklist item in the Safety Center
- Submit a free-text answer in the AI Practice Arena and read the score breakdown
- Switch to an admin account and show the analytics dashboard

Where this goes next

- Real backend + database (Firebase/Supabase) for cross-device progress
- Genuine LLM-backed scoring for the email analyzer and practice arena
- Real authentication (OAuth or hashed email/password)
- PDF/Excel export of a learner's progress report

- Multi-language support
- Push/email “scam of the day” reminders
- CI/CD with GitHub Actions for lint + auto-deploy
- Docker containerization once a backend exists

CONCLUSION

Meaningful security education, zero infrastructure

- A single well-structured HTML file can deliver interactive, case-based cybersecurity learning
- Practical, offline tools give learners something to use immediately, not just study
- Data-driven design (scam cases and checklists as plain arrays) keeps content easy to extend
- A clear path exists to a full-stack version with real auth, persistence, and AI-backed feedback